

TEJASWI REDDY VALLAPU

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PROFESSIONAL SUMMARY

Results-driven **AI/ML Engineer** with **4+ years** of experience designing and shipping production-grade **Machine Learning** and **Generative AI** systems for top-tier **financial services** and **insurance** organizations. Proven expertise in **LLM fine-tuning**, **RAG**, **agentic AI**, **Deep Learning**, **NLP**, and **Computer Vision**, with a track record of measurable business impact including **89% summarization accuracy**, **94% diagnostic accuracy**, and **70%+ faster** document retrieval. Skilled in architecting and deploying scalable **MLOps** pipelines on **AWS**, **Azure**, and **GCP**, with strong cross-functional collaboration across data science, product, and engineering teams.

TECHNICAL SKILLS

GenAI, LLM & Agentic AI: GPT/LLaMA/Claude/Mistral/Gemini fine-tuning, RAG, Prompt Engineering, embeddings & semantic search, LangChain, LlamaIndex, Pinecone, Weaviate, FAISS, Chroma, Milvus, multi-agent orchestration, GenAIOps, conversational AI/chatbot development

Deep Learning & Frameworks: ANN, CNN, RNN/LSTM, Transformers, GANs, Diffusion Models, Reinforcement Learning, TensorFlow, PyTorch, Keras, Scikit-learn, Hugging Face, ONNX

MLOps, Cloud & Big Data: MLflow, Kubeflow, Docker, Kubernetes, Jenkins, GitHub Actions, GitLab CI, FastAPI, Flask, TorchServe, Triton, model versioning/monitoring/drift detection, AWS SageMaker, Azure ML/OpenAI, GCP Vertex AI, Spark, Hadoop, Kafka, Airflow, Snowflake, Redshift

NLP & Computer Vision: Text classification, summarization, tokenization, sentiment analysis, NER, BERT/BART, object detection, image segmentation, OCR, medical image analysis

Data Science & ML Foundations: Linear Algebra, Probability, Statistics, Regression, Decision Trees, Random Forest, SVM, XGBoost, LightGBM, model evaluation (precision, recall, F1, AUC-ROC)

Architecture, Databases & Tools: Solution architecture, REST/gRPC API design, microservices, Postgres, MySQL, MongoDB, Cassandra, Git/GitHub/GitLab, Jupyter, experiment tracking (Weights & Biases), A/B testing

Languages: Python, Java, C++, Scala, R, SQL, Bash/Shell

PROFESSIONAL EXPERIENCE

AI/ML Engineer @ JPMorgan Chase & Co.

Jul 2025 – Present

- Architected a **RAG** pipeline using **LangChain** and **Pinecone** over a fine-tuned **LLaMA** model, cutting document-search time from 18 minutes to **under 3**.
- Fine-tuned **GPT** and **Mistral** models with **LoRA/PEFT** on 40K+ filings, lifting summarization accuracy from 71% to **89%** and saving **25+ hours/week**.
- Led design and delivery of a **multi-agent orchestration** layer with **LlamaIndex**, cutting trade-break reconciliation turnaround from 2 days to **4 hours**.
- Deployed a **GenAIOps** pipeline on **AWS SageMaker** with **MLflow** and **Kubernetes**, catching **3 production regressions** pre-release.
- Served transformer inference via **FastAPI/gRPC** on **Triton**, sustaining **1,200+ requests/second** at low latency.

AI/ML Engineer Intern @ MetLife *(graduate internship, relocated to US for Master's)*

Dec 2024 – Apr 2025

- Developed a **BERT**-based **NLP** pipeline for claims classification and **NER**, raising F1 score from 0.74 to **0.91** and cutting manual claims review by **35%**.
- Trained **CNN/CRNN** models in **PyTorch** with **EasyOCR/Tesseract**, cutting manual data-entry volume by **60%**.
- Implemented an **XGBoost/LightGBM** ensemble for fraud detection, lifting recall by **17 points** at equal false-positive rate.
- Set up **Weights & Biases** tracking and **A/B testing**, cutting the model-iteration cycle from 5 days to **2**.

AI/ML Engineer (Associate Software Engineer) @ Accenture

Jan 2023 – Jun 2023

- Validated architecture for a **conversational AI** chatbot serving **50,000+** monthly users, preventing two production incidents.
- Built a **Spark/Kafka/Airflow** ETL pipeline feeding **Azure ML** and **Azure OpenAI**, cutting batch runtime from 6 hours to **2.5 hours** and reducing nightly compute spend by **30%**.
- Partnered cross-functionally to formalize a **microservices/REST API** design standard adopted by **4 project teams**.

AI/ML Engineer @ LTIMindtree

Dec 2020 – Dec 2022

- Engineered **Spark/Hadoop** pipelines orchestrated with **Airflow**, loading **2TB+** of daily data into **Snowflake** and **Redshift**.
- Developed **CNN**-based object-detection and segmentation models in **TensorFlow/Keras**, raising diagnostic accuracy from 82% to **94%**.
- Containerized **15+ models** with **Docker/Kubernetes** and built **Jenkins/GitLab CI** pipelines, cutting deployment time to **under 2 hours**.
- Designed **Postgres/MongoDB/Cassandra** schemas for feature stores, cutting average query time by **45%** and supporting real-time scoring for **5 downstream ML models**.
- Delivered **regression/SVM/random-forest** forecasting models and mentored 2 engineers, improving forecast accuracy by **11%**.

EDUCATION

University of North Texas — Master of Computer and Information Sciences Aug 2023 – May 2025

Denton, TX — Machine Learning, Deep Learning, NLP, Distributed Systems, Cloud Computing

Jawaharlal Nehru Technological University — Bachelor of Information Technology Jun 2018 – Jul 2022